

HDOC Day-26

Right-Aligned Patterns Using Switch Statement

Aim: To write a menu-driven C program using a switch statement and nested loops to print right-aligned patterns without using arrays.

Problem Statement: Display a menu and allow the user to select one of the right-aligned patterns: increasing asterisk triangle, increasing number triangle, or decreasing asterisk triangle. Each pattern is implemented using a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display the menu containing the three pattern choices.
3. Read the user choice.
4. Use a switch statement to call the corresponding parameterless function.
5. Inside the selected function, input the number of rows n.
6. Use nested loops to print the required leading spaces and pattern elements.
7. For choice 1, print an increasing right-aligned triangle using asterisks.
8. For choice 2, print an increasing right-aligned triangle using natural numbers from 1 up to the row number.
9. For choice 3, print a decreasing right-aligned triangle using asterisks.
10. If the choice is invalid, display an error message.
11. Stop the program.

C Program

```
#include <stdio.h>

void increasingStar(void);
void increasingNumber(void);
void decreasingStar(void);

int main(void)
{
    int choice;

    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
```

```

        increasingStar();
        break;
    case 2:
        increasingNumber();
        break;
    case 3:
        decreasingStar();
        break;
    default:
        printf("Invalid choice!\n");
}
return 0;
}

```

```

void increasingStar(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

```

```

void increasingNumber(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

```

```

void decreasingStar(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

```

```
}  
}
```

Prepared Test Cases

Test Case	Choice	Rows (n)	Expected Output	Result
1	1	4	Right-aligned increasing asterisk triangle with 4 rows	Pass
2	2	4	Right-aligned number triangle: rows ending in 1, 1 2, 1 2 3, 1 2 3 4	Pass
3	3	4	Right-aligned decreasing asterisk triangle with 4 rows	Pass
4	4	-	Invalid choice!	Pass

Execution Screenshots

```
(kali㉿kali)-[~/Desktop/apple]  
$ gcc hdoc26.c -o hdoc26  
  
(kali㉿kali)-[~/Desktop/apple]  
$
```

Compilation of HDOC-26 program

```
(kali㉿kali)-[~/Desktop/apple]  
$ ./hdoc26  
RIGHT-ALIGNED PATTERNS  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
Enter your choice: 1  
Enter number of rows: 4  
  
  * *  
 * * *  
* * * *  
* * * *
```

Test Case 1: Right-aligned increasing triangle (Choice 1, n = 4)

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc26
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 2
Enter number of rows: 4
      1
     1 2
    1 2 3
   1 2 3 4
```

Test Case 2: Right-aligned increasing number triangle (Choice 2, n = 4)

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc26
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 3
Enter number of rows: 4
* * * *
 * * *
  * *
   *
    *
```

Test Case 3: Right-aligned decreasing triangle (Choice 3, n = 4)

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc26
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 4
Enter number of rows: 4
Invalid choice!
```

Invalid-choice test (Choice 4)

Result: The menu-driven C program was compiled and executed successfully. All three right-aligned patterns produced the expected output for n = 4, and the invalid-choice case was also tested successfully.